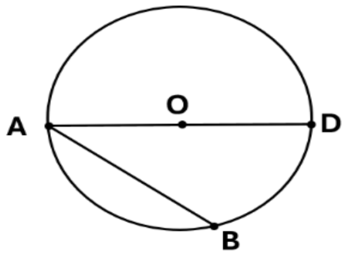
BAL BHARATI PUBLIC SCHOOL
ANNUAL EXAMINATION (2025-26)
CLASS - IX
SUBJECT-MATHEMATICS
SET - A

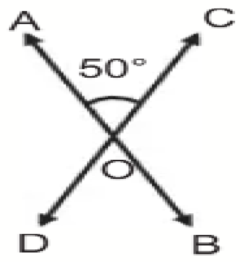
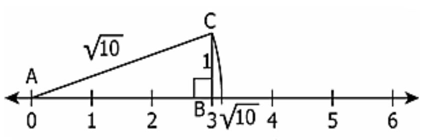
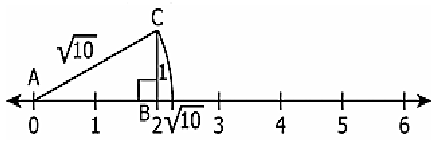
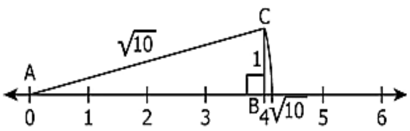
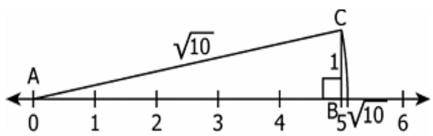
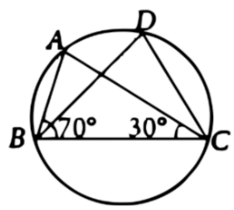
TIME: 3Hrs **Maximum Marks: 80**

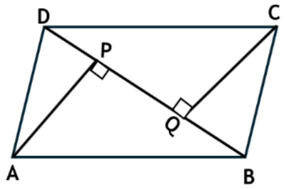
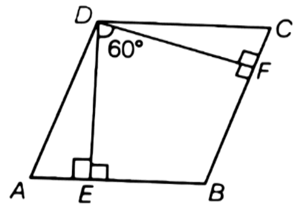
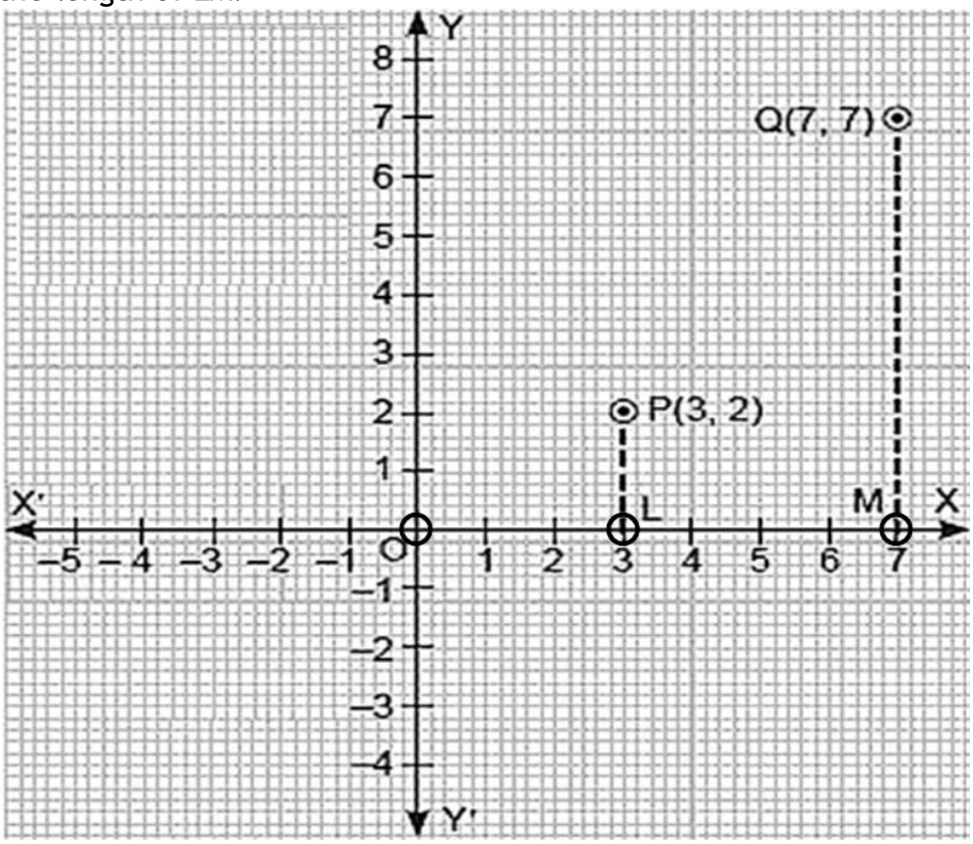
General Instructions

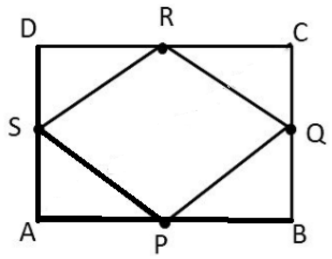
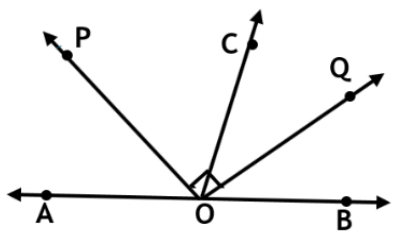
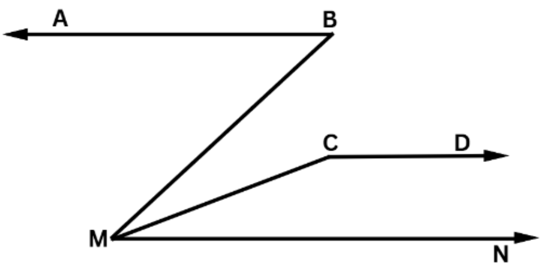
i. *This question paper contains 38 questions. All Questions are compulsory.*
ii. *This Question Paper is divided into 5 Sections A, B, C, D and E.*
iii. *In Section A, Question numbers 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.*
iv. *In Section B, Question numbers 21-25 are very short answer (VSA) type questions, carrying 02 marks each.*
v. *In Section C, Question numbers 26-31 are short answer (SA) type questions, carrying 03 marks each.*
vi. *In Section D, Question numbers 32-35 are long answer (LA) type questions, carrying 05 marks each.*
vii. *In Section E, Question numbers 36-38 are case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.*
viii. *There is no overall choice. However, an internal choice in 2 questions of Section B, 2 questions of Section C and 2 questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.*
ix. *Take* $\pi = \frac{22}{7}$ *wherever required if not stated.*
x. *Use of calculators is not allowed.*

	SECTION - A **Section A consists of 20 questions of 1 mark each.**	
Q1	Perpendicular distance of the point P$(-3, 8)$ from y-axis is: (A) 3 units (B) 8 units (C) 5 units (D) 9 units	1
Q2	The value of $(-12)^3 + (7)^3 + (5)^3$ is: (A) 420 (B) -420 (C) 1260 (D) -1260	1
Q3	The length of each side of an equilateral triangle having an area of $9\sqrt{3}$ cm^2 is: (A) 8 cm (B) 36 cm (C) 4 cm (D) 6 cm	1

Q4	Select the expression which is a polynomial: (A) $x^{-2} + x^{-1} + 5$ (B) $x^2 + 5\sqrt{x} + 7$ (C) $\frac{1}{x} + 7$ (D) $\frac{\sqrt{x^3}}{\sqrt{x}} + 7$	1
Q5	If the volume and surface area of a sphere are numerically equal, then its radius is: (A) 1 unit (B) 2 units (C) 3 units (D) 4 units	1
Q6	Select the statement which is false: (A) A straight line may be drawn from any one point to any other point. (B) A terminated line cannot be produced indefinitely. (C) A circle can be drawn with any centre and any radius. (D) All right angles are equals to one another.	1
Q7	A rational number lying between $\sqrt{2}$ and $\sqrt{3}$ is: (A) $\frac{\sqrt{2}+\sqrt{3}}{2}$ (B) 1.5 (C) $\sqrt{2}+\sqrt{3}$ (D) 2.5	1
Q8	AD is the diameter of a circle and AB is its chord. If AD = 10 cm and AB = 6 cm. Then the distance of chord AB from the centre of the circle is: (A) 8 cm (B) 6 cm (C) 4 cm (D) 2 cm 	1
Q9	It is given that $\Delta ABC \cong \Delta FDE$ and AB = 5 cm, $\angle B = 40^\circ$ and $\angle A = 80^\circ$. Then which of the following is true? (A) DF = 5 cm, $\angle F = 60^\circ$ (B) DE = 5 cm, $\angle D = 40^\circ$. (C) DE = 5 cm, $\angle F = 80^\circ$ (D) DF = 5 cm, $\angle E = 60^\circ$	1
Q10	Ordinate of all the points on x axis is: (A) 1 (B) -1 (C) 0 (D) Any number	1
Q11	The width of each of the five continuous classes in a frequency distribution is 5 and the lower class limit of the lowest class is 10. The upper class -limit of the highest class is: (A) 35 (B) 40 (C) 15 (D) 45	1
Q12	"Things which coincide with one another are equal to one another." The above statement is a/an: (A) Axiom (B) Definition (C) Theorem (D) Postulate	1
Q13	Select the option that expresses $5y + 8x = 8(x+y) - 9$ in the form $ax + by + c = 0$. (A) $0.x + 3y + 9 = 0$ (B) $0.x + 3y - 9 = 0$ (C) $3x - 3y - 9 = 0$ (D) $0.x - 3y - 9 = 0$	1
Q14	The largest sphere is cut off from a cube of side 6cm. The surface area of the sphere will be: (A) $144\pi\ cm^2$ (B) $108\pi\ cm^2$ (C) $36\pi\ cm^2$ (D) $324\pi\ cm^2$	1

Q15	ABCD is a parallelogram and E is the mid -point of BC .DE and AB when produced meet at F. Then AF is: (A) $\frac{3}{2}$ AB (B) $\frac{1}{2}$ AB (C) 3AB (D) 2AB	1
Q16	In the given figure, $\angle AOC = 50^{\circ}$, then $\angle AOD + \angle COB$ is equal to: (A) 100° (B) 140° (C) 260° (D) 130°  	1
Q17	Which of the following represents $\sqrt{10}$ on a number line? (A)  (B)  (C)  (D)  	1
Q18	The polynomial having $x - 1$ as a factor is: (A) $x^4 + x^3 + 5x^2 + x + 1$ (B) $x^4 + x^3 + x^2 + x + 1$ (C) $x^3 + 2x^2 - 2x + 1$ (D) $x^3 - x^2 + x - 1$	1
Q19	**Assertion (A):** In the given figure, $\angle ABC = 70^{\circ}$ and $\angle ACB = 30^{\circ}$. Then, $\angle BDC = 80^{\circ}$ **Reason (R):** Angles in the same segment of a circle are equal. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true and (R) is not the correct explanation of (A). (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.  	1
Q20	**Assertion (A):** The graph of the linear equation $2x - 9y = 18$ cuts x-axis at the point (2,0). **Reason (R):** Every point lying on the graph of $2x - 9y = 18$ is a solution of the equation. (A)Both (A) and (R) are true and (R) is the correct explanation of (A). (B)Both (A) and (R) are true and (R) is not the correct explanation of (A). (C) (A) is true but (R) is false (D) (A) is false but (R) is true	1
	SECTION B **Section B consists of 5 questions of 2 marks each.**	
Q21	Factorise: $x^2 - 1 - 2a - a^2$.	2

Q22	ABCD is a parallelogram and AP and CQ are perpendiculars from vertices A and C on diagonal BD. Prove that AP = CQ.  **OR** In the given fig, the angle between two altitudes of a parallelogram ABCD through the vertex of an obtuse $\angle D$ of the parallelogram is 60°. Find $\angle A$ and $\angle B$.  	2
Q23	The radius of a spherical balloon becomes $\frac{3}{4}^{th}$ of its original radius when air is released. Find the ratio of the volume of air present in the original balloon to that after deflation. **OR** A dome is in the shape of a hemisphere. The total cost of whitewashing it from outside is ₹15,400. If the rate of whitewashing is ₹200 per square metre, find the radius of the dome.	2
Q24	Observe the given figure and answer the questions given below: (a) Find the coordinates of point L. (b) What will be the coordinate of point R such that OMQR forms a square? (c) Find the mirror image of point P along x-axis. (d) Find the length of LM.  	2
Q25	Find the product of: $\left(p-\frac{1}{p}\right)\left(p+\frac{1}{p}\right)\left(p^2+\frac{1}{p^2}\right)\left(p^4+\frac{1}{p^4}\right)$	2

	<u>SECTION C</u> **Section C consists of 6 questions of 3 marks each.**	
Q26	ABCD is a rectangle and P, Q, R and S are mid-points of the sides AB, BC, CD and DA respectively. Show that the quadrilateral PQRS is a rhombus. 	3
Q27	If a + b + c = 15 and $a^2 + b^2 + c^2 = 83$, find the value of $a^3 + b^3 + c^3 - 3abc$.	3
Q28	If the non-parallel sides of a trapezium are equal, prove that it is cyclic.	3
Q29	Find the value of p and q when $\frac{\sqrt[4]{2} \times \sqrt{27}}{(32)^{\frac{1}{4}} \times (3)^{\frac{1}{2}}} = 2^p \times 3^q$ **OR** If A = $0.\overline{23}$ and B = $(\sqrt{5} + \sqrt{4})(\sqrt{5} - \sqrt{4})$ then, solve and express A + B in the form $\frac{p}{q}$,where p and q are integers and $q \neq 0$.	3
Q30	In the given figure, OP bisects ∠AOC, OQ bisects ∠BOC and OP ⊥ OQ. Show that the points A, O and B are collinear.  **OR** In the given figure, ∠ABM = 55°, ∠BMC = 20°, ∠CMN = 35° and ∠MCD = 145°. Prove that AB || CD. 	3
Q31	Find the value of a and b if the points P(b,0) and Q(a,b) are the solutions of the linear equation y = 2x − 8. Hence, find the value of k if Q(a, b) lies on 3x + y − 2k = 0.	3
	Section - D **Section D consists of 4 questions of 5 marks each**	
Q32	Prove that the angle subtended by an arc at the centre is twice the angle subtended by it at any point on the remaining part of the circle. Hence prove that angle in a semicircle is a right angle.	5

Q33 If $(x - \frac{1}{x}) = 4$, then evaluate $(x^2 + \frac{1}{x^2})$ and $(x^3 - \frac{1}{x^3})$. **5**

OR

If the polynomial $4x^3 - 16x^2 + ax + 7$ is exactly divisible by $x-1$, then find the value of a. Hence factorise the polynomial.

Q34 If $a = \frac{1}{3+\sqrt{11}}$ and $b = \frac{1}{a}$, then find $a^2 - b^2$. **5**

Q35 The marks obtained (out of 60) by a class of 80 students are given below : **5**

Marks	10-20	20-30	30-40	40-50	50-60
Number of students	5	15	23	20	17

Represent the above data with the help of a histogram and a frequency polygon on the same graph.

OR

Following is the frequency distribution

Class Interval	100-150	150-200	200-300	300-500	300-800
Frequency	60	100	100	80	180

Draw a histogram to represent the above data.

SECTION E

Section E consists of 3 case study-based questions of 4 marks each.

Q36 International kite festival in Gujarat also known as Uttarayan is one of the biggest festival celebrated in Gujarat.

On the day of Uttarayan, Suresh, a kite maker, received an order to make following triangular kites:

Kite 1: A triangular kite with sides 10 units, 10 units, and 12 units.

Kite 2: A triangular kite with sides in the ratio 3:4:5 and a semi-perimeter of 12 units.

Answer the following questions using Heron's Formula:

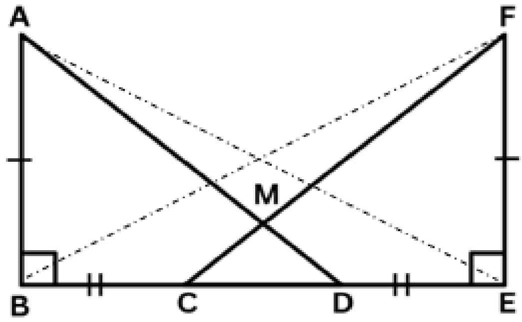
(i) Find the semi-perimeter of Kite 1. **1**

(ii) Find the area of Kite 1. **1**

(iii) (a) Find the area of Kite 2. **2**

OR

(iii) (b) If the cost of paper used to make a kite is ₹1.50 per sq. unit, then find the cost of making 200 kites of kind 1. **2**

Q37	Modern bridges are designed using triangular frameworks because triangles provide maximum strength and stability. Consider a pedestrian bridge where two equal vertical pillars AB and FE are erected on the ground. Two iron rods AD and FC are fixed on the base such that they intersect at point M and BC = DE, forming triangular structures.  Answer the following question on the basis of the given information.  (i) Prove that BD = CE. (ii) Prove that Δ ABD $\cong$ Δ FEC. (iii) (a) Prove Δ MCD is an isosceles triangle. **OR** (iii) (b) Prove that Δ AED $\cong$ Δ FBC.	 1 1 2 2
Q38	After harvesting, a farmer stores wheat in an open field by piling it in the shape of a cone. To protect the wheat from rain, the heap is to be covered with a canvas. The diameter of one such conical heap is 1.4 m and its vertical height is 2.4 m. Based on the above situation, answer the following questions: (i) Find the slant height of the conical heap so formed.  (ii) Find the area of the canvas required to cover the heap. (iii) (a) If $\frac{1}{8}^{th}$ of the volume of the conical heap is occupied by air, find the volume occupied by the wheat grains in the heap. **OR** (iii) (b) A new heap is formed such that both the radius and the height are doubled. Find the ratio of the curved surface area of the new heap to that of the original heap.	 1 1 2 2